

Miaomiao Chen

Sr. Machine Learning Engineer, Ph.D.

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SUMMARY

Machine Learning Engineer with 7+ years in **MLOps, Large-Scale Data Analysis, and Agentic AI**. Expert in building production-grade LLM systems (RAG, Post-train) and scalable Kubernetes microservices for pharmaceutical needs. **China National Chess Champion**, bringing strategic depth, resilience, and rigorous mathematical thinking to complex engineering challenges.

EDUCATION

M.S. in Computer Science (Machine Learning Track), Georgia Institute of Technology	2024 - present (Remote)
Ph.D. in Chemistry (Computational Modeling), University of Virginia, VA	2019 - 2024
B.S. in Bioscience, ShanghaiTech University, China	2015 - 2019

TECHNICAL SKILLS

Programming & Infra: Python, SQL, Spark, Nextjs ; Docker, Kubernetes, vLLM

Machine Learning: Transformers(GPT, BERT, VIT, HSTU), Agents (LangChain), PyTorch, Reinforcement Learning (RL)

EXPERIENCE

- Sr. Research Scientist (Machine Learning Focus)** | AstraZeneca PLC | MD Jul 2024 – Present
- Led the development of domain-specialized **AI Agents** using **LangChain, RAG, and MonetDB**; integrated **analytical tools** for multi-cohort comparisons and statistical testing, accelerating cancer patient data interpretation by **3x**.
 - Architected an unsupervised **Transformer-based anomaly detector** for high-dimensional instrument data, designing a novel **"run-load gap" embedding** to capture complex operational drift superior to standard time-series methods.
 - Engineered a full-stack QC analytical platform (**FastAPI, Next.js**) deployed on **Kubernetes**; integrated into the core pipeline to automate statistical profiling and visualization, enabling users to rapidly assess sample integrity and instrument health.
 - Engineered a **Dense Passage Retrieval (DPR)** system by fine-tuning **BGE embeddings** via **contrastive learning with in-batch negatives**; optimized semantic alignment on 7k+ proteomics Q&A pairs to power high-precision RAG queries.
 - Fine-tuned encoder-decoder **Foundation Models** on clinical trial data to forecast drug response rates; achieved a **20% accuracy uplift**, directly accelerating the ranking and prioritization of high-potential therapeutic candidates.
- Graduate Researcher (Computational Modeling)** | University of Virginia | VA 2019 – 2024
- Engineered a sparse proteomics imputation pipeline benchmarking **NMF vs. KNN**; validated via masked RMSE to prove NMF's superior latent structure preservation, resolving key limitations in existing software.
 - Engineered an end-to-end predictive modeling pipeline (**Random Forest, XGBoost, LSTM**) for high-dimensional drug candidate prioritization, achieving **85.3% accuracy** and a **20% wet-lab validation hit rate**.
 - Architected data analysis workflows and A/B testing frameworks to identify novel inflammation mechanisms; validated computational insights via in-vivo mouse models, resulting in a **first-author publication in PNAS**.

SELECTED PROJECTS

- Krono: Voice-Activated AI News Agent** 🗣️ 🔄 | Python, LangGraph, vLLM, GRPO 2025 – Present
- Architected a **multi-agent system** using **LangGraph** and **SenseVoice ASR** for intent detection and tool routing, enabling natural, interruptible voice interactions with persistent conversational context.
 - Post-trained Qwen-1.5B** using **verl (Ray overrides) + GRPO (Group Relative Policy Optimization)** and deployed via **vLLM**, reducing inference costs by **50%** while maintaining high classification accuracy.
 - Designed a real-time threading architecture to orchestrate audio streams and **RAG retrieval**, achieving **<100ms latency** while reducing user information lookup time by **~70%** compared to manual search.
- GPT-2 Reproduction & Optimization** 🧠 🛠️ | PyTorch, Transformers, CUDA 2024 – Present
- Built and trained a custom **GPT-2 (124M)** style model from scratch (12 layers, 384-dim), integrating **FlashAttention** to significantly accelerate training throughput and optimize GPU memory utilization.
 - Implemented core Transformer operators (e.g., **Softmax, RoPE, Causal Attention**) from first principles following the **Stanford CS336** curriculum, mastering low-level matrix arithmetic (MatMul) and computation graphs.
 - Validated robustness on the **HellaSwag** benchmark (26% accuracy) after 3,800+ training steps.
 - Implemented **QLoRA** for memory-efficient fine-tuning on multilingual datasets, demonstrating expertise in adapting large models on constrained hardware resources.
- Generative Retrieval Video RecSys (HummingbirdRec)** 🐦 | PyTorch, HSTU, Transformers 2024 – Present
- Architected a SOTA **Generative Recommender** based on **HSTU (Hierarchical Sequential Transduction Unit)** for UGC short-video feeds, outperforming strong baselines (SASRec) by **+11% in HR@10 and NDCG@10**.
 - Engineered a custom **RoPE** mechanism injecting **timestamp and duration-aware signals**, capturing fine-grained temporal dynamics to achieve a **15% NDCG uplift**.
 - Designed a **self-adaptive label normalization** strategy based on **watch-ratio** (playtime/duration), reducing implicit feedback noise and driving an additional **~3% gain** on the **KuaiRec** benchmark.